Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Potometer	1			1		1
		(ii)		(Leafy shoot must be) cut underwater. (1) {Apparatus / description of} must be assembled underwater / potometer flooded with water. (1)	2			2		2
		(iii)		**Any two for one mark** Temperature / light <u>intensity</u> / humidity		1		1		1
	(b)	(i)	I	157 / 157.08 / 157.1 (2 marks) Award 1 mark for (3.14 or π) x 0.5^2 x 200 628 / 628.3 (use of diameter instead of radius)		2		2	2	2
			II	49.1 (3 marks) **If incorrect award 2 marks for** 49 / 49.06 / 49.09 (correct answer but not to 1 dp) **If incorrect award 1 mark for** 157 (answer from (b)(i) / 192 x 60 Allow ecf from (b)(i)		3		3	3	3
		(ii)		**Any three (x1) from** A. As fan distance increases the <u>rate</u> of water uptake decreases (1) B. As distance increases transpiration (rate) decreases (1) C. Water vapour accumulates / humidity increases / diffusion shells remain outside of the leaf (1) OWTTE D. Decreases the steepness of the {diffusion / water potential} gradient (1) ignore concentration Accept reverse argument for each marking point		3		3		3

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			**Any one (x1) from** Some water taken up may have been used in {photosynthesis / hydrolysis reactions / increased turgor} (1) Some water lost may have been produced in {respiration / condensation reactions} (1) Some water lost may have been due to cuticular evaporation (1)		1		1		1
				Question 4 total	**3**	**10**	**0**	**13**	**5**	**13**